ESP32-CAM Text-to-Speech and OCR Glasses Project Summary

Overview:
The project aims to develop a smart wearable assistive device that captures text from the real world and converts it to audible speech output, helping visually impaired individuals and also aiding travelers in reading foreign text such as signs and menus. The system uses an ESP32-CAM module for image capture and wireless transmission, with PaddleOCR running on a local server for text recognition.

Hardware Components:
1. ESP32-CAM module with OV2640 camera (main processing and Wi-Fi transmission)
2. 3.7V Li-ion battery (power source)
3. Step-up converter (3.7V to 5V boost converter)
4. PAM8403 audio amplifier module (for driving earphones or small speaker)
5. Audio jack or small speaker (output for speech)
6. Jumper wires and optional breadboard for prototyping

System Workflow:
1. The ESP32-CAM captures an image from the environment.
2. The image is preprocessed — resolution is reduced and converted to grayscale.
3. The processed image is sent to a local computer via Wi-Fi, where PaddleOCR extracts text from the image.
4. The recognized text is sent back to the ESP32 via Wi-Fi.
5. The ESP32 forwards the text to the PAM8403 amplifier, which drives the speaker or headphone output, converting the text into audible speech.

Power Notes:
- ESP32-CAM can operate at 3.7V but may be unstable; using a step-up converter to provide a stable 5V is recommended.
- The PAM8403 amplifier accepts 5V input and drives small 3W speakers or earphones.

Alternate Implementation – Live Stream OCR Method:
In this approach, instead of capturing and sending still images, the ESP32-CAM acts as a live streaming camera (webcam). The local machine (laptop or server) runs PaddleOCR on the video feed in real-time. This allows text recognition from the continuous live stream instead of static images, offering faster feedback and dynamic reading capability.
Steps:
1. Configure the ESP32-CAM to broadcast video via Wi-Fi (e.g., using ESP32-CAM MJPEG stream).
2. On the laptop, open the live stream URL using OpenCV.
3. Feed each frame to PaddleOCR for real-time text detection.
4. Display or process the recognized text as needed (can be integrated with text-to-speech on the PC).